

Linear dependence on the number of simultaneous Riesz selectors

Full answer to the open problem after Corollary 4.8 of arXiv:2405.18235

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Status. `candidate_full_solution_likely_valid_needs_human_review.`

Abstract

Bownik proves that blocks of size $O(m^2 B / \varepsilon^2)$ suffice for a simultaneous Riesz selector from m unit-norm Bessel families of common bound B , and asks whether the quadratic dependence on m is optimal. We prove that $16mB^2 / \varepsilon^2$ suffices. The proof applies the paper's simultaneous Bessel-selector theorem once to the normalized families and all their Naimark complements. A cyclic construction gives a matching linear lower bound for fixed $B > 1 + \varepsilon$. Thus the answer is no, and for example at $B = 2$ the optimal dependence on m is $\Theta(m)$.

1 Source problem and proof intuition

For each $j \in [m]$, let $(u_i^{(j)})_{i \in I}$ be a unit-norm Bessel sequence of bound $B > 1$, possibly in a different Hilbert space. Given pairwise disjoint blocks $(J_k)_{k \in K}$, a selector is a set $J \subset \bigcup_k J_k$ with $|J \cap J_k| = 1$ for every k .

Corollary 4.8 of [1] obtains one selector for which all m subfamilies have Riesz bounds $1 - \varepsilon, 1 + \varepsilon$, provided

$$|J_k| \geq C \frac{m^2 B}{\varepsilon^2}.$$

The source then asks whether the quadratic m -dependence is asymptotically optimal; see Figure 1.

The direct idea is to normalize each family to Bessel bound one and add its Naimark complement. An upper Bessel estimate on the normalized original gives the desired upper Riesz bound. The same estimate on its complement gives the desired lower Riesz bound. Crucially, the squared-norm parameters of each original/complement pair add to one. Across m pairs, the simultaneous selector error therefore depends on m , not m^2 .

2 Two source ingredients

We isolate the exact consequences of Corollary 3.4 and Lemma 4.3 of [1].

Lemma 1 (Simultaneous Bessel selector). *For ℓ in a finite or countable set L , let $(w_i^{(\ell)})_{i \in I}$ be Bessel with bound one and suppose $\|w_i^{(\ell)}\|^2 \leq d_\ell$, where $\sum_{\ell \in L} d_\ell < \infty$. If $|J_k| \geq r$ for every selector block, there is a selector J such that every $(w_i^{(\ell)})_{i \in J}$ has Bessel bound at most*

$$d_\ell + \frac{1}{r} + 2\sqrt{\frac{\sum_{q \in L} d_q}{r}}.$$

Lemma 2 (Naimark-complement criterion). *If $(a_i)_{i \in I}$ is Bessel with bound one and $\|a_i\|^2 = \eta \in (0, 1)$, there is a Bessel-one family $(v_i)_{i \in I}$ with $\|v_i\|^2 = 1 - \eta$ such that, for every $J \subset I$,*

$$(a_i)_{i \in J} \text{ has lower Riesz bound } \delta \iff (v_i)_{i \in J} \text{ has Bessel bound } 1 - \delta.$$

3 Linear upper bound

Theorem 1 (Linear simultaneous selector). *Let $m \geq 1$, $B > 1$, and $0 < \varepsilon < 1$. Suppose that for every $j \in [m]$, $(u_i^{(j)})_{i \in I}$ is a unit-norm Bessel sequence of bound B . If the pairwise disjoint selector blocks satisfy*

$$|J_k| \geq R := \left\lceil \frac{16mB^2}{\varepsilon^2} \right\rceil, \quad k \in K,$$

then there is a selector J such that every $(u_i^{(j)})_{i \in J}$ is a Riesz sequence with bounds $1 - \varepsilon$ and $1 + \varepsilon$.

Proof. Put $\eta = B^{-1}$ and $a_i^{(j)} = B^{-1/2}u_i^{(j)}$. Each $a^{(j)}$ is Bessel with bound one and $\|a_i^{(j)}\|^2 = \eta$. Lemma 2 supplies a complement family $v^{(j)}$ with squared norms $1 - \eta$.

Apply Lemma 1 once to all $2m$ families

$$a^{(1)}, \dots, a^{(m)}, v^{(1)}, \dots, v^{(m)}.$$

Their norm parameters sum to

$$m\eta + m(1 - \eta) = m.$$

Hence one selector J simultaneously gives Bessel bounds

$$\eta + E_R \quad \text{for every } a^{(j)}|_J, \quad 1 - \eta + E_R \quad \text{for every } v^{(j)}|_J,$$

where $E_R = R^{-1} + 2\sqrt{m/R}$. The definition of R gives

$$2\sqrt{m/R} \leq \frac{\varepsilon}{2B}, \quad \frac{1}{R} \leq \frac{\varepsilon}{2B},$$

the second inequality using $m \geq 1$, $B > 1$, and $0 < \varepsilon < 1$. Thus $E_R \leq \eta\varepsilon$.

The first Bessel estimate is the upper Riesz bound $\eta(1 + \varepsilon)$ for $a^{(j)}|_J$. The complement estimate is

$$1 - \eta + \eta\varepsilon = 1 - \eta(1 - \varepsilon),$$

so Lemma 2 gives lower Riesz bound $\eta(1 - \varepsilon)$ for $a^{(j)}|_J$. Scaling by $B^{1/2}$ multiplies both bounds by $B = \eta^{-1}$, proving the claim. $\square$

For fixed B and ε , Theorem 1 is linear in m , already answering the source question negatively. Combined with the source estimate, it also gives the joint sufficient bound

$$O(\varepsilon^{-2} \min\{mB^2, m^2B\}).$$

4 The linear exponent is sharp

Theorem 2 (Cyclic lower bound). *Fix $0 < \varepsilon < 1$ and $B > 1 + \varepsilon$. For every $m \geq 1$ and every $1 \leq r \leq m$, there are m unit-norm Bessel families of bound at most B and two selector blocks of size r for which no simultaneous selector has Riesz bounds $1 - \varepsilon, 1 + \varepsilon$.*

Proof. Choose α with $\varepsilon < \alpha \leq \min\{1, B - 1\}$. Let $(e_p)_{p \in \mathbb{Z}/r\mathbb{Z}}$ and $(f_q)_{q \in \mathbb{Z}/r\mathbb{Z}}$ be orthonormal and mutually orthogonal. Use two blocks

$$J_0 = \{x_p : p \in \mathbb{Z}/r\mathbb{Z}\}, \quad J_1 = \{y_q : q \in \mathbb{Z}/r\mathbb{Z}\}.$$

For $j \in \mathbb{Z}/r\mathbb{Z}$, define

$$u_{x_p}^{(j)} = e_p, \quad u_{y_q}^{(j)} = \alpha e_{q+j} + \sqrt{1 - \alpha^2} f_q.$$

If $m > r$, repeat any of these families. For fixed j , both displayed subfamilies are orthonormal, and their cross-Gram matrix is α times a permutation matrix. The full Gram matrix therefore has eigenvalues $1 - \alpha$ and $1 + \alpha$; the family is unit norm and has Bessel bound $1 + \alpha \leq B$.

Any selector is a pair $\{x_p, y_q\}$. In the family with $j = p - q \pmod{r}$, its two vectors have inner product α , so their Gram eigenvalues are $1 - \alpha$ and $1 + \alpha$. Since $\alpha > \varepsilon$, this selector fails the requested bounds. $\square$

Corollary 1 (Exact exponent of m). *For every fixed $0 < \varepsilon < 1$ and $B > 1 + \varepsilon$, the optimal universal block size grows as $\Theta(m)$. In particular this holds for $B = 2$.*

Verification and novelty. An exact verifier checked 4,050 scalar instances of the upper-bound constants and all 22,140 cyclic selector pairs through size 40. Four run indexes and bounded exact-title, exact-question, simultaneous-selector, and Naimark-complement searches found no prior resolution. Because the proof is a short combination of two results in [1], expert novelty review is advised.

Limitation. The asymptotic- m problem is fully answered. The optimal joint dependence on m, B, ε is not determined: the new proof pays B^2 , while the source's proof pays $m^2 B$.

References

- [1] M. Bownik, *Selector form of Weaver's conjecture, Feichtinger's conjecture, and frame sparsification*, arXiv:2405.18235 (2024).

It is not clear whether the bound in Theorem 4.6 is asymptotically optimal as a function of the collection of parameters $\{\epsilon_j\}_{j \in N}$. For simplicity, assume that all ϵ_j are all equal, which by the Blaschke condition $\sum_{j \in N} (1 - \epsilon_j) < \infty$, necessarily forces the set N to be finite. Corollary 4.8 is an extension of [14, Theorem 3.8] in two ways. It yields a simultaneous selector for multiple Bessel sequences. At the same it gives an improved, asymptotically optimal bound for a single Bessel family in terms of Riesz sequence tightness parameter $\varepsilon > 0$, which was conjectured in [14, Theorem 3.8].

Corollary 4.8. *There exists a universal constant $C > 0$ such that the following holds. Let $m \in \mathbb{N}$, $B > 1$, and $\varepsilon > 0$. Suppose that for each $j \in [m]$, $\{u_i^{(j)}\}_{i \in I}$ is a Bessel sequence with bound B and*

$$\|u_i^{(j)}\|^2 = 1 \quad \text{for all } i \in I, j \in [m].$$

Let $\{J_k\}_{k \in K}$ be a collection of disjoint subsets of I with

$$(4.11) \quad \#|J_k| \geq C \frac{m^2 B}{\varepsilon^2} \quad \text{for all } k \in K.$$

Then, there exists a selector $J \subset \bigcup_k J_k$ satisfying (3.2) such that for all $j \in [m]$, $\{u_i^{(j)}\}_{i \in J}$ is a Riesz sequence with bounds $1 - \varepsilon$ and $1 + \varepsilon$.

Note that the size of sets J_k in Corollary 4.8 is proportional to the Bessel bound B , and inversely proportional to ε^2 , which controls the tightness of resulting Riesz sequence. These bounds are asymptotically optimal as either $B \rightarrow \infty$ or $\varepsilon \rightarrow 0$, see [13, 44]. However, it not clear at all whether the size of sets J_k needs to grow proportionally to m^2 . The paving result of Ravichandran and Srivastava, Corollary 3.8, suggests that a linear growth might be possible.

Problem. *Is the bound in (4.11) asymptotically optimal bound as $m \rightarrow \infty$?*

Figure 1: Source page 26: Corollary 4.8 and the asymptotic- m problem.